9.2 Calcium Ions

Ca^{2+} play two critical roles in the physiology of cardiac muscle. Their influx through slow Ca channels accounts for the prolonged Plateau-phase & absolute refactory period that enable cardiac muscle to function properly. Ca^{2+} also combine with regulatory protein troponin - tropomyosin complex
↳ This complex removes the inhibition that prevents the ↳ heads of the myosin molecules from forming cross bridges with the active sites on actin that provide the power stroke of contraction ↳ virtually identical to skeletal muscle →

Approx 20% of Ca^{2+} ~~of the~~ required for contraction is supplied by the influx of Ca^{2+} during the Plateau phase.

The Sarcoplasmic reticulum releases the remaining Ca^{2+}

Comparative Rates of Conduction System Firing

The pattern of prepotential or spontaneous depolarization → following rapid depolirizat-ion & repolarization ~ as in the SA Node

SA Node is the pacemaker, → it reaches threshold faster than any other ~~component~~ ~~conductive~~ cells in heart component of the ↗ conduction